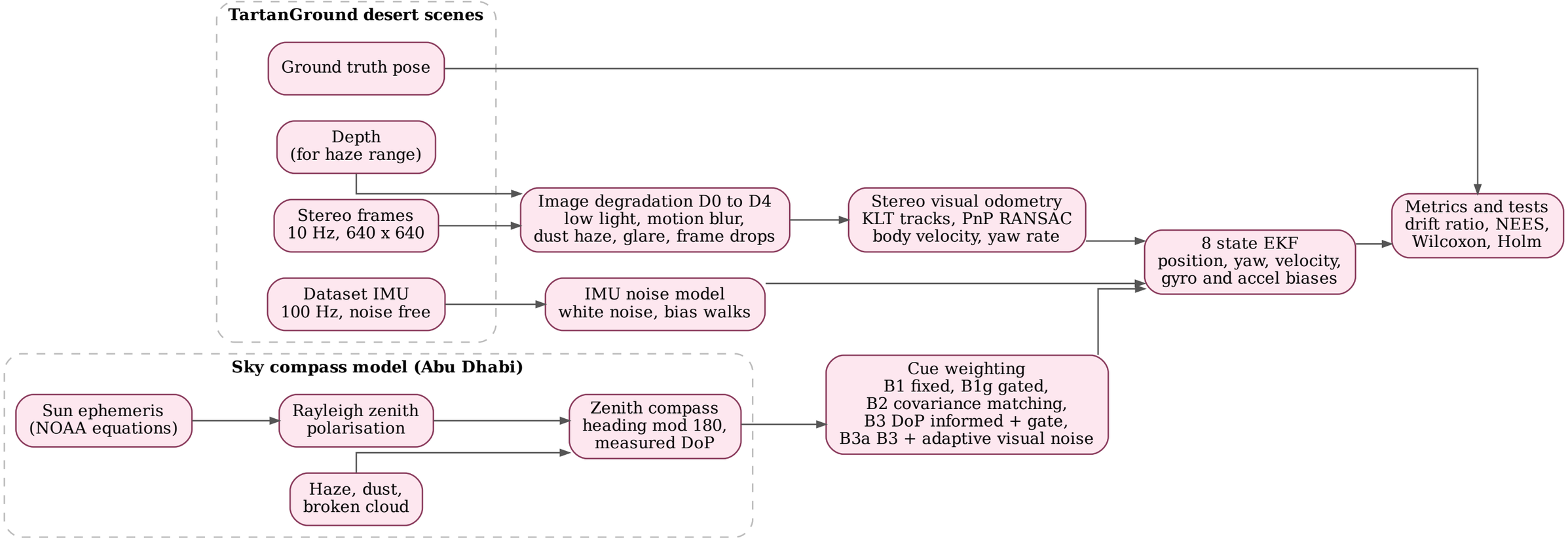
TartanGround desert scenes
Ground truth pose
Depth
(for haze range)
Stereo frames
10 Hz, 640 x 640
Dataset IMU
100 Hz, noise free
Image degradation D0 to D4
low light, motion blur,
dust haze, glare, frame drops
IMU noise model
white noise, bias walks
Stereo visual odometry
KLT tracks, PnP RANSAC
body velocity, yaw rate
8 state EKF
position, yaw, velocity,
gyro and accel biases
Metrics and tests
drift ratio, NEES,
Wilcoxon, Holm
Sky compass model (Abu Dhabi)
Sun ephemeris
(NOAA equations)
Rayleigh zenith
polarisation
Haze, dust,
broken cloud
Zenith compass
heading mod 180,
measured DoP
Cue weighting
B1 fixed, B1g gated,
B2 covariance matching,
B3 DoP informed + gate,
B3a B3 + adaptive visual noise